ZS-U2

Reheating Dynamics: Dual-Channel Decay, Three-Timescale Hierarchy, Gravitational Wave Spectrum, and ε-SM Coupling

Kenny Kang

Version 1.0 — March 2026
Theme: Early Universe [ZS-U] | Paper 2 of 8

Verification: 24/24 PASS | Zero New U2 Fit Parameters | A_s Normalized

§0. Abstract

We derive the reheating dynamics of the Z-Spin ε-field from the base action of ZS-F1 v1.0, introducing **no new fit parameters** beyond A = 35/437 and the Planck normalization A_s. The ε-field couples to Standard Model particles through the conformal factor $\Omega^2 = 1 + \mathbf{A}\varepsilon^2$, generating an effective coupling y_eff = **A**/(1+**A**) = 35/472 ≈ 0.074. The three-timescale hierarchy T_osc (4.7 t_P) ≪ t_decay (1,912 t_P) ≪ t_Hubble (267,410 t_P) implies dual-channel reheating: ~406 coherent oscillations during which perturbative conformal decay and parametric resonance operate simultaneously.

The conformal decay rate $\Gamma = y_{eff}^2 m^3_{eff}/(8\pi M^2_P)$, independently derived from the Z-Spin action, yields Γ/H_end ≈ 140 ≫ 1 (instant reheating). The reheating temperature **T_reh = (30 ρ_end/(π^2 g_*))^{1/4} ≈ 2.55 × 10^{15} GeV** follows from the instant reheating limit, where ρ_end = 1.5 λ_inf V_E(ε_end) M^4_P includes the kinetic correction. The decay branching ratios predict 85% gluons, 7% W-bosons, and 8% B-bosons via the SM trace anomaly.

We compute the gravitational wave spectrum from three sources: the inflationary tensor background ($\Omega_{GW} h^2 \approx 5 \times 10^{-18}$, detectable by LiteBIRD at ~9σ and DECIGO at 500× sensitivity), parametric resonance GW (peak at ~12.6 THz, unobservable), and thermal plasma GW (peak at ~67 MHz, unobservable). ZS-U2 sharpens the ZS-U1 v1.0 prediction r = 0.0089 by eliminating the reheating-temperature uncertainty band. r_ZS/r_Starobinsky = 2.63, enabling LiteBIRD to distinguish Z-Spin from Starobinsky at 6σ.

Keywords: reheating, trace anomaly, parametric resonance, gravitational waves, LiteBIRD, DECIGO, branching ratios, conformal coupling, instant reheating

§0.1 Epistemic Status Legend

Status	Definition

PROVEN	Exact mathematical identity/theorem under declared definitions.
STANDARD	Established result in QFT/cosmology textbooks.
DERIVED	Quantitative consequence from Z-Spin axioms + standard physics.
VERIFIED	Numerically checked (verification suite provided).
TESTABLE	Quantitative prediction with explicit falsification condition.
HYPOTHESIS	Structural pattern without completed derivation chain.
OBSERVATION	Empirical pattern identified; upgrade to DERIVED pending derivation.
NON-CLAIM	Quantity NOT derived; honest acknowledgment of limitation.
WITHDRAWN	Previously claimed result found incorrect; formally retracted.

§1. Introduction and Motivation

The Z-Spin action $S = \int d^4x \sqrt{(-g)}\, [\frac{1}{2}(1+\mathbf{A}\varepsilon^2)R - \frac{1}{2}(\nabla \varepsilon)^2 - V(\varepsilon)]$ predicts inflation (ZS-U1 v1.0), but the post-inflationary universe must thermalize. ZS-U2 derives the complete reheating process from this action, introducing zero new fit parameters beyond A = 35/437, the A_s normalization, and standard SM gauge couplings.

The conformal factor $\Omega^2 = 1 + \mathbf{A}\varepsilon^2$ is the sole bridge between the ε-field and Standard Model matter. Through the trace anomaly of the SM energy-momentum tensor, this coupling generates decay channels to all gauge bosons. The single geometric parameter **A = 35/437** determines the coupling strength, decay rate, branching ratios, reheating temperature, and gravitational wave spectrum.

§2. Conformal Coupling to the Standard Model

In the Einstein frame, the ε-field acquires an effective coupling to SM fields through the conformal transformation. The effective Yukawa-like coupling is:

$$y_eff = A/(1+A) = 35/472 \approx 0.0742 \qquad (1)$$

This is NOT a free parameter; it is algebraically determined by **A = 35/437** (ZS-F2 v1.0).

[STATUS: DERIVED] y_eff from base action conformal factor. No new parameters.

§3. Three-Timescale Hierarchy

Three characteristic timescales govern the post-inflationary dynamics:

Table 1. Three-timescale hierarchy.

Timescale	Expression	Value	Physical meaning

T_osc	2π/(m_eff M_P)	4.7 t_P	One ε-oscillation period
t_decay	8π M²_P/(y²_eff m³_eff)	1,912 t_P	Conformal decay time
t_Hubble	1/H_end	267,410 t_P	Expansion timescale

The hierarchy T_osc ≪ t_decay ≪ t_Hubble implies that the ε-field executes approximately **N_osc ≈ 406** coherent oscillations before decaying, while the Hubble friction acts only at the longest timescale.

Γ/H ≈ 140 ≫ 1 → instant reheating

[STATUS: DERIVED + VERIFIED] All timescales independently derived from A, m_eff, λ_inf.

§4. Decay Branching Ratios

The ε-field decays to SM gauge bosons through the conformal trace anomaly. The partial width to each gauge sector is proportional to $|b_0|^2$ N_i α^2_i, where b_0 is the 1-loop β-function coefficient:

Table 2. Decay branching ratios from trace-anomaly coupling.

Channel	b_0 (full SM)	N_gauge	α(M_P)	BR	Sector
ε → gg (gluons)	7	8	0.0198	85.0%	SU(3)
ε → WW	19/6	3	0.0206	7.1%	SU(2)
ε → BB	−41/6	1	0.0175	7.9%	U(1)

[STATUS: DERIVED] Standard SM trace-anomaly with Z-Spin conformal coupling.

§5. Reheating Temperature and Dual-Channel Dynamics

The conformal decay rate is independently derived from the Z-Spin action:

$$\Gamma_pert = y^2_eff\, m^3_eff / (8\pi\, M^2_P) \qquad (2)$$

where m_eff = √(m²_eff_norm) × M_P and m²_eff_norm = d²V_E/d$\tilde{\varphi}$²|_{ε=1} ≈ 1.79 is computed from the Einstein-frame potential (ZS-U1 v1.0). This yields Γ/H_end ≈ 140 ≫ 1, placing the system firmly in the instant reheating regime.

In the instant reheating limit (Γ ≫ H), the reheating temperature is determined by the energy density at inflation end, not by the standard perturbative formula T ~ √(ΓM_P):

$$T_reh = (30\, \rho_end / (\pi^2\, g_*))^{1/4} \approx 2.55 \times 10^{15}\ GeV \qquad (3)$$

where $\rho_{end} = 1.5 \times \lambda_{inf} \times V_E(\varepsilon_{end}) \times M^4_P$ includes the kinetic correction factor 1.5. This is well above the leptogenesis threshold ($>10^9$ GeV, ZS-S5 v1.0) and below the GUT scale ($\sim 2\times10^{16}$ GeV). Dual-channel reheating operates: perturbative conformal decay dominates globally, while parametric resonance ($q = y_{eff} \approx 0.074$, narrow regime) provides additional early-time energy transfer for approximately 35 oscillations.

Note: The standard perturbative formula $T = (90/(\pi^2 g_*))^{1/4} \sqrt{(\Gamma M_P)}$ yields $\sim 3\times10^{16}$ GeV, which overestimates by ~12× because it assumes a matter-dominated reheating phase. In the instant regime ($\Gamma/H = 140$), the ρ_{end} formula (Eq. 3) is the physically correct expression.

[STATUS: DERIVED] T_{reh} from ρ_{end}. All inputs independently computed from $A = 35/437$.

§6. Gravitational Wave Spectrum from Reheating

6.1 Source Classification

Table 3. GW source classification.

GW Source	Peak $\Omega_{GW} h^2$	Peak frequency (today)	Detectable?
Inflationary tensor (ZS-U1 v1.0)	5.1×10^{-18}	Flat (10^{-18}–67.5 MHz)	✓ YES
Preheating parametric res.	$\sim 7 \times 10^{-18}$	~12.6 THz	✗ NO
Thermal plasma	$\sim 2 \times 10^{-13}$	~67 MHz	✗ NO

[STATUS: DERIVED] All from independently computed inputs (A, r, T_{reh}).

6.2 Inflationary Tensor Background

$$\Omega_{GW}^{inf} h^2 = (P_t/24) \times \Omega_{rad} h^2 \times \langle T^2 \rangle \times R(g_*) \approx 5.1 \times 10^{-18} \quad (4)$$

High-frequency cutoff: $f_{max} \approx 67.5$ MHz. Because reheating is instant ($\Gamma/H = 140$), there is no suppression of high-frequency modes below f_{max}.

$r_{ZS}/r_{Starobinsky} = 0.0089/0.00339 = 2.63$. LiteBIRD ($\sigma_r \approx 0.001$) can distinguish Z-Spin from Starobinsky at **6σ** significance. CMB-S4 ($\sigma_r \approx 5\times10^{-4}$) provides complementary confirmation.

[STATUS: DERIVED + TESTABLE] Parameter-free from ZS-U1 v1.0 + ZS-U2.

6.3 Preheating GW (Unobservable)

$f_{peak} \approx 12.6$ THz (mid-infrared $\lambda \sim 24$ μm). No GW detector operates at this frequency. This is a NON-RESULT honestly reported.

6.4 Thermal Plasma GW (Unobservable)

$f_{peak}^{th} \approx 67$ MHz. The thermal plasma GW peak falls in an unobservable gap between LIGO (10–100 Hz) and no planned detector.

§7. ε-Higgs Effective Field Theory

At the Z_2 attractor ($\varepsilon_2 = \cos(\arg z^*) \approx 0.8066$), the Higgs potential rescales as $V_H^{E} = V_H/(1+A)^2 \approx 0.857\, V_H$. This is shape-invariant: the VEV $v = 246$ GeV is independent of the overall potential scale.

Radiative portal coupling: $\xi_{eff} \sim A^2/(16\pi^2) \approx 4\times10^{-5}$ (negligible). Thermal decoupling: $T_{reh}/m_\varepsilon \approx 7.8 \times 10^{-4} \rightarrow$ Boltzmann suppression.

[STATUS: DERIVED] Higgs parameters (m_H, v) are NON-CLAIM (§9).

§8. Falsification Conditions

Table 4. Falsification conditions for ZS-U2.

ID	Condition	Timeline	Detector
FU2-1	$r \neq 0.0089 \pm 0.002$	~2032	LiteBIRD
FU2-2	BR deviates from SM trace anomaly by >3σ	~2040	Collider + BBN
FU2-GW1	$\Omega_{GW} h^2$ at 0.1 Hz differs from 5×10^{-18} by >3×	~2040s	DECIGO
FU2-GW2	Preheating GW detected at $f < 10^6$ Hz	~2040s	Any GW detector

§9. Non-Claims (Honest Limitations)

NC1: Higgs mass $m_H = 125.25$ GeV is NOT derived from A. NON-CLAIM.

NC2: Higgs VEV $v = 246$ GeV is NOT derived from A. NON-CLAIM.

NC3: Reheating GW detectability at $f > 10^6$ Hz is a NON-RESULT.

NC4: Sector interpretation ($I_h \leftrightarrow$ oscillation, $O_h \leftrightarrow$ lattice) is HYPOTHESIS.

§10. Discussion and Conclusions

ZS-U2 establishes the complete reheating dynamics of Z-Spin Cosmology from **A = 35/437** with no new U2 fit parameters. The conformal decay rate $\Gamma = y^2_{eff} m^3_{eff}/(8\pi M^2_P)$ is independently derived, giving $\Gamma/H = 140$ (instant reheating). The reheating temperature $T_{reh} = 2.55 \times 10^{15}$ GeV follows from the ρ_{end}-based instant reheating formula. The sole detectable primordial GW is the inflationary tensor background, measurable by LiteBIRD (~9σ), CMB-S4 (~18σ), and DECIGO (500×). Reheating GW are honestly assessed as unobservable. $r_{ZS}/r_{Starobinsky} = 2.63$ provides a decisive discriminator.

§11. Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy/SciPy, with fully independent derivation chain) is publicly available.

Appendix A. Verification Suite Summary

24 tests across 6 categories: independent derivations (5), timescale hierarchy (4), reheating dynamics (4), branching ratios (4), GW spectrum (5), cross-paper consistency (2). All quantities independently computed from A = 35/437 — no hardcoded intermediate values. 100% pass rate.

Appendix B. T_reh Formula Selection

In the instant reheating regime ($\Gamma/H \approx 140 \gg 1$), the ρ_{end}-based formula $T_{reh} = (30\rho_{end}/(\pi^2 g_*))^{1/4}$ is the physically correct expression. The standard perturbative formula $T \sim (90/(\pi^2 g_*))^{1/4} \sqrt{(\Gamma M_P)}$ assumes a matter-dominated reheating phase ($\Gamma \ll H$) and overestimates T_{reh} by ~12× in the instant regime. The verification suite explicitly tests this distinction (test C4).

Appendix C. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Independent Derivations	5	5/0	y_{eff}, m^2_{eff}, H_{end}, $\Gamma/H=140$, T_{reh}
[B] Timescale Hierarchy	4	4/0	$T_{osc}=4.7$, $t_{decay}=1912$, $t_{Hub}=267410\ t_P$
[C] Reheating Dynamics	4	4/0	T_{reh} bounds, $q<1$, formula consistency
[D] Branching Ratios	4	4/0	BR(gg)=85%, sum=100%
[E] GW Spectrum	5	5/0	$\Omega_{GW}=5.1\times10^{-18}$, $r_{ratio}=2.63$
[F] Cross-Paper	2	2/0	A×437=35 exact, derivation chain

TOTAL: 24/24 PASS — 100% pass rate

Cross-Reference Table

Result	Status	Dependencies
y_eff = A/(1+A) = 35/472	DERIVED	ZS-F1 v1.0 (base action), ZS-F2 v1.0 (A)
$\Gamma = y^2m^3/(8\pi M^2)$, $\Gamma/H=140$	DERIVED+VERIFIED	ZS-U1 v1.0 (m_eff, ε_end), independent computation
T_reh = 2.55×10^{15} GeV	DERIVED	ρ_end instant reheating ($\Gamma/H \gg 1$)
BR(gg/WW/BB) = 85/7/8%	DERIVED	SM trace anomaly + ZS-S1 v1.0
Ω_GW $h^2 = 5.1\times10^{-18}$	DERIVED+TESTABLE	ZS-U1 v1.0 (r=0.0089)
r_ZS/r_Staro = 2.63	DERIVED	N_* = 59.5 (ZS-U1 v1.0)
Reheating GW unobservable	NON-CLAIM	f_peak = 12.6 THz

References

[1] K. Kang, "The Z-Spin Action & U(1) Completion," ZS-F1 v1.0 (2026).

[2] K. Kang, "Geometric Impedance: A = 35/437," ZS-F2 v1.0 (2026).

[3] K. Kang, "ε-Field Inflation," ZS-U1 v1.0 (2026).

[4] K. Kang, "Gauge Coupling Unification," ZS-S1 v1.0 (2026).

[5] K. Kang, "Resonant Leptogenesis Framework," ZS-S5 v1.0 (2026).

[6] K. Kang, "Z-Sim Forward Simulator," ZS-T3 v1.0 (2026).

[7] Kofman, L., Linde, A., & Starobinsky, A.A., Phys. Rev. D 56, 3258 (1997).

[8] Fujii, Y. & Maeda, K., The Scalar-Tensor Theory of Gravitation, Cambridge (2003).

[9] Tristram, M. et al., Phys. Rev. D 105, 083524 (2022). $r < 0.032$.

[10] LiteBIRD Collaboration, PTEP 2023, 042F01 (2023). $\delta r \approx 0.001$, launch ~JFY2032.

[11] CMB-S4 Collaboration, ApJ 926, 54 (2022). $\sigma(r) \approx 5\times10^{-4}$.

[12] Kawamura, S. et al., CQG 28, 094011 (2011). DECIGO.

[13] Planck Collaboration, A&A 641, A10 (2020).

Version History

v1.0 (March 2026): Initial public release. Reheating dynamics with independently derived T_{reh} from ρ_{end} formula (Eq. 3), conformal decay rate Γ (Eq. 2), branching ratios, GW spectrum, ε-Higgs EFT, dual-channel dynamics, T_{reh} formula selection (Appendix B). All intermediate values independently computed — no hardcoded quantities. 24/24 tests across 6 categories. (Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0)